Green Home Score & Retrofit Recommendation Engine

TEAM 103

1. Problem Understanding

Homeowners often do not know how energy-efficient their homes are or which improvements would make the biggest difference. Insulation, heating systems, windows, and renewable energy all affect energy use, bills, and carbon emissions. Choosing the right improvements can be difficult without a clear comparison of their likely benefits.

Our goal is to build a tool that gives each home a Green Home Score from 0 to 100, explains the main reasons behind that score, and recommends suitable improvements. It will also estimate possible changes in energy use, carbon emissions, and property value.

Rather than giving users just a number, the system aims to explain what could be improved first and why.

2. Our Approach

We have created code to generate a synthetic dataset of 5,000 properties and have built the main parts of the solution around these records. The dataset gives us a consistent way to test the scoring and recommendation pipeline without using confidential customer information.

2.1 Green Home Score

Each property receives a weighted score based on seven components, using the weights specified in the problem statement.

#	Component	Weight
1	Energy efficiency	25%
2	Energy use per square metre	20%
3	Insulation	15%
4	Heating system	15%
5	Windows and glazing	10%
6	Renewable energy	10%
7	Smart energy management	5%

Each component is converted into a sub-score out of 100. The final score is calculated as:

$$S = \sum_{i=1}^{7} w_i s_i$$

Here, s_i is a component sub-score and w_i is its weight as a fraction. The weights sum to 1. The sub-score mappings will be documented, and improving a component must never reduce its score.

Homes are placed into the following categories:

- **80–100:** Highly efficient
- **65–79:** Good
- **50–64:** Moderate
- **30–49:** Poor
- **0–29:** Very poor

Energy use is considered per square metre, rather than by total consumption alone. This helps avoid marking a larger home as inefficient simply because it uses more energy.

2.2 Retrofit Recommendations

Homes classified as Poor or Very poor will be highlighted for improvement. The system will rank properties by their estimated potential to reduce annual purchased energy.

For each property, it will suggest feasible improvements, assign a suitability level (High, Medium, or Low), estimate potential benefits, and explain why an option may be useful. Possible improvements include loft insulation, double glazing, heat pumps, solar panels, and smart thermostats.

The order of improvements also matters. For example, improving insulation before installing a heat pump may be more suitable for some homes. Savings from several upgrades will not simply be added together, since improvements can affect one another.

3. Energy and Carbon Estimates

For each proposed improvement, the system estimates annual electricity and gas changes separately, in kWh/year. This is important because a heat pump, for example, may reduce gas consumption while increasing electricity consumption.

The emissions factors used in our calculations are:

- **Electricity:** 0.207 $kgCO_2e/kWh$
- **Gas:** 0.183 $kgCO_2e/kWh$

Estimated carbon savings are calculated from the changes in each fuel's consumption and its corresponding emissions factor. Both reductions and increases must be included, so an increase in electricity use is not incorrectly counted as a saving.

Where used by the implementation, the heat pump calculation uses a coefficient of performance (COP) of 3.2, while annual solar PV generation or savings is capped at 2,500 kWh/year. These are modelling assumptions, not guarantees of actual property performance.

3.1 Property Value and AI Explanations

We train a gradient boosted regressor (LightGBM) on regional property attributes, floor area, and efficiency scores to predict hedonic property valuations. Outputs are reported as an indicative prediction interval (e.g., +2% to +5%), explicitly communicating market correlation rather than guaranteed causation.

The AI assistant explains scores, recommendations, and calculated results in everyday language. Numerical values must come from the scoring and calculation systems rather

than being invented by the language model.

For a what-if scenario, the system first reruns the relevant calculations and then asks the AI to explain the updated results. This keeps the explanation tied to the actual outputs.

To provide personalized continuity, a persistent Customer Memory module tracks homeowner budget constraints and previously rejected retrofits.

4. System Pipeline

The solution follows this high-level sequential flow:

1. Generate and validate synthetic property data.
2. Calculate component scores and the final score.
3. Assign an efficiency category to each property.
4. Recommend and rank suitable retrofits.
5. Estimate electricity, gas, and carbon changes, as well as property-value changes where implemented.
6. Generate explanations and updated what-if results.
7. Present the results in a user-facing dashboard.

Keeping calculations separate from AI explanations makes the numerical results easier to check and helps reduce unsupported claims.

5. Testing and Evaluation Plan

We will assess the system using the following checks:

- **Data quality:** Check missing values, invalid ranges, duplicates, and consistency in the generated property records.
- **Scoring:** Check the seven weights, category boundaries, weighted calculations, and the rule that better component conditions cannot lower the corresponding sub-score.
- **Recommendations:** Check suitability, prioritisation, and whether interactions between improvements are considered.
- **Energy and carbon:** Verify units, calculation direction, separate electricity and gas changes, and the emissions factors.
- **Predictive models:** Benchmark our LightGBM property value regressor against a baseline median predictor using held-out RMSE, MAE, and R^2 on the customer-isolated test set.
- **AI explanations:** Compare explanations with actual calculation outputs. Targets are greater than 90% groundedness, greater than 90% factual accuracy, and fewer than 5% unsupported claims. These are evaluation targets, not assumed results.
- **What-if scenarios:** Confirm that changing property details triggers the required recalculations and that the explanation reflects the updated outputs.

Since our initial dataset is synthetic, its evaluation cannot alone establish real-world accuracy. Further validation using suitable real-world data would be needed before treating the estimates as proven household outcomes.